Estimating Suppressed Electricity Demand in a Rationed Grid: A Disequilibrium Approach for Nigeria

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ABSTRACT

Electricity demand in chronically supply-constrained grids is systematically mismeasured. Where supply is rationed and tariffs are administratively set, observed consumption equals the smaller of demand and supply and understates true demand by a variable, unobserved margin, biasing the forecasts on which capacity investment depends. Existing estimates of this "suppressed demand" for Nigeria and comparable markets fix the correction from consumption to demand by assumption before estimation. This paper instead recovers it, applying the Fair–Jaffee/Maddala–Nelson disequilibrium framework (in which observed quantity is the minimum of separately specified, jointly estimated demand and supply equations) to a panel of 154 distribution-company-quarters (11 DISCOs, 2022Q4–2026Q1) built from Nigerian regulatory filings. Demand is identified by state-level formal economic activity; supply by administratively allocated contracted capacity and national plant availability; the demand- or supply-constrained status of each observation is recovered from the likelihood rather than assumed. The existing connected network is estimated to draw 5.34 GW on average against 3.43 GW delivered: a suppressed demand gap of 1.91 GW, 56 per cent above current delivery. The estimate holds out of sample and is insensitive to the reference year and to the April 2024 tariff change, but it rests on the three DISCOs whose quarters reveal demand: excluding any one of them moves the gap to between 0.52 and 1.48 GW, and a plausible negative correlation between demand and supply shocks would lower it by up to two-fifths.

1.0 Introduction

Nigeria has roughly 220 million people, an installed generation capacity on paper of about 13 GW, and, on an average day, delivers under 4 GW to the grid. The shortfall is not disputed. What no one can state with confidence is a more useful number: how much electricity the country's distribution network would actually take if the power were there to take. That figure (not installed capacity, not consumption, but latent, constrained demand) is what an investor sizing a generation project, or a regulator pricing new capacity, actually needs. It is also, by construction, unobservable.

The reason is a measurement trap specific to rationed markets. When a grid is chronically short and tariffs are set by regulation rather than cleared by price, the electricity that changes hands is whichever is smaller (what buyers want, or what the system can supply) and never reveals which (Fair and Jaffee, 1972). Consumption data, the natural starting point for any demand estimate, therefore understates true demand by an amount that is not only large but variable: it shifts period to period as the binding constraint moves between generation, transmission, and distribution. A demand forecast built on that series inherits the bias silently. In a sector where the size of the investment programme depends directly on the projected demand, underestimating demand produces underinvestment, which reproduces the shortage, the mechanism by which a measurement error becomes a self-perpetuating physical one.

Existing estimates of Nigeria's "suppressed demand" do not escape this trap so much as assume their way around it. The most recent (Ugbehe et al., 2025) inflates historical consumption by a fixed formula for outages, idle capacity, and losses; a close Ghanaian analogue (Dramani et al., 2024) builds its suppressed-demand series from researcher-assigned weights before modelling it. The only prior disequilibrium treatment of Nigerian electricity (Ojameruaye, 1988) fixes a single reliability index for the entire study period and never estimates the supply side at all. In each, the correction is an input, chosen before estimation; none lets the data determine how large the gap is, or how it moves.

This paper does. Demand and supply are estimated as two separately specified equations, jointly fit by maximum likelihood, with observed delivery modelled as their minimum (Fair and Jaffee, 1972; Maddala and Nelson, 1974). The method assumes no reliability index, no market-clearing price, and, critically, no prior labelling of which quarters were demand- or supply-constrained; that classification falls out of the estimation itself. Applied to 154

distribution-company-quarters (11 DISCOs, 2022Q4–2026Q1) built from Nigeria's regulator's own filings, with demand identified by state-level formal economic activity and supply by contracted capacity and plant availability, the model estimates that Nigeria's existing, connected network would draw 5.34 GW against the 3.43 GW it actually receives: a suppressed demand gap of 1.91 GW, roughly 56 per cent above current delivery. The estimate holds out of sample; its dependence on the three DISCOs that identify demand, and on the independence assumption, is reported in Sections 6.4 and 6.5.

The problem this method solves is no longer Nigeria's alone. Since 2024, the same mathematical structure (quantity rationed below what the market would otherwise clear) has surfaced at the opposite end of the development spectrum, as AI and data-centre load outruns the grid's physical ability to connect it. US interconnection queues held roughly 2,061 GW of active generation and storage capacity at the end of 2025, around one and a half times the country's installed generating base, with a median wait from interconnection request to commercial operation exceeding five years for projects completed that year (Rand et al., 2026). The rationing there is not in the wholesale price, which does respond to scarcity, but one stage upstream, in the queue for connection itself: a market with its own latent demand (load seeking a connection), its own supply (connection capacity delivered per year), and, again, only their minimum observed. It is the same estimation problem in a richer setting, and it sharpens the case for getting the method right where the stakes are longest-standing. Nigeria serves as that case here, with the wider application taken up in Section 8.

2.0 Literature Review

2.1 Estimating electricity demand when the market does not clear

The problem of inferring demand from consumption in a rationed market is well understood in principle and routinely sidestepped in practice. Three approaches recur in the electricity literature for developing economies, distinguished less by statistical sophistication than by how much they require the analyst to assume before estimation begins.

The most direct is deterministic gross-up: observed consumption is inflated by adjustment factors intended to restore what would have been consumed absent constraint. Ugbehe et al. (2025) produce the most recent national estimate for Nigeria this way, scaling historical consumption by factors for capacity utilisation, access rate, outages, and transmission-and-distribution losses before extrapolating forward. The appeal is transparency; the cost is that each factor is a judgement fixed in advance, and the resulting figure can be no more defensible than those inputs. Nothing in the method lets the data revise them.

A second approach adds statistical machinery downstream but leaves the core problem upstream. Dramani et al. (2024), estimating suppressed demand for Ghana over 2000–2020, model a suppressed-demand series against transmission losses, informal-economy size, and temperature using a quantile ARDL specification, then forecast it forward under alternative socioeconomic scenarios. The quantile approach is a genuine advance: it captures how the drivers of suppressed demand act differently across the distribution. But suppressed demand is not observed: the series being explained must first be constructed from observed consumption and a set of adjustment assumptions before any regression can run. The sophistication is applied to explaining that series, not to identifying it, the same boundary the gross-up approach meets, reached one step later.

The third approach is closest to ours, and the only prior application of disequilibrium econometrics to Nigerian electricity: Ojameruaye (1988). Written in the pre-privatisation NEPA era, it recognises explicitly that observed sales are the short side of a non-clearing market and that demand is therefore latent, the correct starting point. But to recover demand, it assumes "a reliability index of 0.95 for the system throughout the period of analysis" (Ojameruaye, 1988, p. 373) and divides observed sales by it. By the authors' own account, "unlike most of the existing econometric disequilibrium models, our model does not attempt to

estimate both the supply and demand sides of the market" (p. 376). It reaches for the right framework, then assumes the two things the framework exists to recover: the size of the constraint, and its variation over time.

The common thread is visible. In each case the correction from consumption to demand (its magnitude, and how it moves) enters as an input, fixed before the data are consulted. None treats the gap between demand and supply as a quantity the data can identify on their own.

2.2 The disequilibrium framework

The tool that removes this assumption is neither new nor untested. Fair and Jaffee (1972) set out the econometric problem directly: "in the absence of an equilibrium condition the ex ante demand and supply quantities cannot in general be equated to the observed quantity traded in the market" (p. 497). Their solution was to model the two quantities as separate equations and the observed quantity as the smaller of the two, estimating both by maximum likelihood without requiring the market to clear. Maddala and Nelson (1974) generalised this into a family of four models, distinguished by how much is known about the direction of price adjustment. In the first (the specification this paper uses) nothing is assumed about price behaviour at all: the model is defined solely by a demand equation, a supply equation, and the condition that observed quantity equals their minimum (Maddala and Nelson, 1974, p. 1013). Regime classification is recovered from the likelihood rather than imposed.

The framework has since been applied across markets where price fails to clear quantity: disequilibrium credit rationing, housing starts under regulatory constraint, and labour markets, among others. Its defining feature (that either buyers or sellers are pushed off their notional schedule whenever the market fails to clear, leaving observed quantity on the "short side") is exactly the condition of a chronically supply-constrained electricity grid, where distribution companies routinely take less than they would draw at the prevailing tariff because the power is not available to take.

The framework carries a known estimation hazard, and one this paper takes seriously. Maddala and Nelson identified it themselves: 'the possibility of several local maxima to the likelihood functions and convergence to a local maximum,' a problem they noted 'need[s] to be investigated further' (1974, p. 1030). A modern, peer-reviewed implementation of all four models now exists (Karapanagiotis, 2024), used in Section 6 as an independent check on the

estimator developed here. That check reproduces precisely the failure the original authors warned of, repeatedly, and on two independently constructed versions of the estimation sample: the comparison package's runs consistently converge to inferior local optima rather than the global one, with the optimisation explicitly reported as non-converged in every instance. This is why the estimation strategy in Section 5 relies on many independent starting points, an analytically derived rather than numerically approximated gradient, and an explicit filter against degenerate solutions, rather than a single optimisation call.

2.3 The gap

Across these literatures, one combination is absent. Suppressed electricity demand has been estimated for developing economies, including Nigeria, but always by fixing the correction from consumption to demand before estimation, through a deterministic gross-up (Ugbehe et al., 2025), a pre-constructed index (Dramani et al., 2024), or an assumed reliability factor (Ojameruaye, 1988). The disequilibrium framework that would let the data determine that correction has been applied widely to other markets, but the one prior attempt to bring it to Nigerian electricity used only its demand half, holding supply and reliability fixed. No study estimates electricity demand in a developing-country grid as a jointly identified latent variable: the demand and supply equations are estimated together, and the demand- versus supply-constrained status of each observation is recovered from the likelihood rather than assumed. That is the gap this paper fills.

Table 1. Prior approaches to estimating suppressed electricity demand.

Study	**Setting**	**Approach**	**Demand-Supply Split**	**Supply side estimated**
Ugbehe et al. (2025)	Nigeria	Deterministic gross-up of consumption	Assumed (fixed factors)	No
Dramani et al. (2024)	Ghana	Quantile ARDL on a constructed index	Assumed (pre-constructed)	No
Ojameruaye (1988)	Nigeria (NEPA era)	Disequilibrium, fixed reliability index	Assumed (R = 0.95, constant)	No

This paper	Nigeria (post-2013)	Joint ML disequilibrium (Maddala–Nelson Model 1)	Estimated from the data	Yes

3.0 Nigeria's post-privatisation electricity market

Nigeria's electricity supply industry was privatised in November 2013, when the state monopoly was unbundled into eleven regional distribution companies (DISCOs), six generation companies, and a single, state-retained transmission company (Transmission Company of Nigeria), the "11-6-1" structure that still defines the sector (Daily Trust, 2025). The eleven DISCOs are territorial monopolies: each holds the exclusive licence to distribute electricity within a defined franchise area, and together they cover the entire country. This structure makes the DISCO-quarter the natural unit of analysis: eleven comparable entities, each facing the same national generation and transmission system but serving distinct territories with different economic and demographic profiles.

Three features of how the sector operates are essential to what follows.

Supply is constrained upstream of the DISCOs. Electricity reaches a DISCO only after passing through generation and the single national transmission grid. Both are chronically limited: against installed generation capacity of roughly 13 GW, average power actually delivered to the grid over the study period was under 4 GW, the shortfall driven by gas supply constraints, plant unavailability, and transmission bottlenecks on the supply side. A DISCO cannot draw more than the system delivers to it, regardless of how much its customers would consume, which is precisely the condition under which observed offtake is the short side of a non-clearing market.

Each DISCO's supply is administratively allocated, not purchased at a clearing price. The volume of power a DISCO is entitled to draw is set through its contracted capacity (a regulated allocation, revised through NERC's tariff-order process), not determined by a bid into a spot market. This matters for identification: because the allocation is set administratively and revised on a regulatory schedule, it shifts a DISCO's available supply without being a response to that DISCO's demand in the same quarter, the exclusion restriction Section 5 relies on.

Tariffs do not clear the market. Retail tariffs are set by the regulator under the Multi-Year Tariff Order and held below cost-reflective levels by government subsidy; as NERC's own chairman has described it, what a customer pays is "a blend of regulation and policy," with the government covering the difference between the regulated and the cost-reflective price (AllAfrica, 2024). Price therefore does not adjust to equate demand and supply. When the grid

is short, the shortfall is resolved by rationing (load-shedding), not by a price rise that clears the market. This is the defining precondition of the disequilibrium framework: with price unable to do the work of allocation, quantity is rationed, and observed consumption reveals the short side rather than true demand.

4.0 Data

4.1 Sources

The panel is built from two primary sources, both official and both public.

The first is the quarterly reports of the Nigerian Electricity Regulatory Commission (NERC), which publishes, for each of the eleven DISCOs, the operational and commercial data the sector runs on: energy received from the grid, contracted capacity, billing and collection, customer counts, and, at the plant level, generation availability. These reports are the source of every supply- and quantity-side variable in the model. Because NERC's tabular formats vary across years, the series were extracted through a purpose-built pipeline combining automated page location with structured extraction, then cleaned and normalised into a consistent DISCO-quarter panel; the construction decisions that materially affect the estimation sample are documented in Section 4.2.

The second is the National Bureau of Statistics' record of state-level Internally Generated Revenue (IGR), the revenue each state government raises from taxes, fees, and levies within its own territory. IGR serves as the model's demand shifter, and the rationale for that choice is set out in Section 5. We use a single NBS-sourced series covering 2016 to 2024, aggregated from constituent states to DISCO franchise areas.

4.2 Data-constrained research design

The estimation window, and several variable choices, are determined by what NERC's public reporting archive makes available and comparable — not by a theoretically preferred horizon or specification. Three constraints shaped the design directly.

The panel begins in 2022Q4 because that is where NERC's reports first provide offtake and contracted capacity in the hourly-average (MWh/h) form the model requires; earlier reports do not carry these variables in a comparable unit. It ends in 2026Q1 because that is NERC's most recent published quarterly report at the time of estimation.

State-level GDP was considered as an alternative demand shifter to IGR and set aside: Nigeria's National Bureau of Statistics published state nominal GDP for 2013–2017 only, covering a subset of states, with no comparable series since. No state-level GDP series exists at the

resolution or currency this panel requires. IGR (2016–2024, all states, single primary source) was adopted for this reason.

Customer counts were tested as a demand-side variable during specification search and excluded, not for lack of explanatory content, but because NERC's shift from "registered" to "active registered" customer reporting between the 2024 and mid-2025 filings introduces a definitional break that cannot be reliably harmonised within the available reports (Section 6.3).

4.3 Panel Construction

The unit of observation is the DISCO-quarter. The estimation sample comprises 154 observations: eleven DISCOs across fourteen quarters, from 2022Q4 to 2026Q1, the full span over which NERC reports offtake and contracted capacity at the hourly-average level (MWh/h) the model requires, the unit at which the supply constraint actually binds. Two construction decisions within this window affect the sample and are stated explicitly.

First, plant availability for four early quarters (2022Q4–2023Q3) was not present in the standard extracted plant tables. For three of them (2022Q4–2023Q2), it was reconstructed from the average available generation capacity stated in each quarter's report narrative, divided by installed capacity; the method was validated by applying it to 2023Q3, where the report separately states the resulting figure (33.31 per cent) directly, and the reconstruction matched to two decimal places. For 2023Q3 itself, an initial automated extraction returned an incorrect availability figure (36.07 per cent), which was identified by cross-checking against the report's own stated value and corrected before estimation. No other variable or quarter required correction.

Second, a definitional break in reported customer counts (NERC's shift from "registered" to "active registered" customers between the 2024 and mid-2025 reports) produces a large, artificial drop in the customer series for most DISCOs at that transition. Because the final specification does not use customer counts (Section 5), this break does not enter the estimated model; it is noted here because it shaped an earlier stage of variable selection and because it affects any customer-based series a reader might construct from the same source.

4.4 Variables and descriptive statistics

The model uses four quantities directly. Observed offtake (Q) is the energy each DISCO draws from the grid, in MWh/h. Contracted capacity (PCC) is the administratively allocated ceiling on that draw. Plant availability (PAF) is the share of national installed generation available in the quarter. State IGR, aggregated to the DISCO, proxies formal economic activity on the demand side. Table 2 reports their distributions.

Table 2. Descriptive statistics for the estimation sample (154 DISCO-quarters).

Variable	Mean	SD	Min	Median	Max
Offtake, Q (MWh/h)	311.9	143.7	57.0	282.8	587.3
Contracted Capacity, PCC (MWh/h)	331.6	148.7	62.2	291.6	611.0
Plant Availability, PAF (%)	37.1	3.3	31.9	37.3	42.6
Offtake ratio, Q/PCC	0.93	0.07	0.70	0.94	1.17

n = 154 DISCO-quarters (11 DISCOs × 14 quarters, 2022Q4–2026Q1).

The offtake ratio, observed draw as a fraction of contracted capacity, is worth isolating, because it is the empirical signature that motivates the entire approach. If DISCOs always took their full contracted capacity, the ratio would sit at or above one throughout, the market would be uniformly supply-constrained, and there would be no demand variation to identify. If they always fell short, it would be uniformly demand-constrained. Neither holds. The ratio averages 0.93 but ranges from 0.70 to 1.17: it exceeds one in six per cent of DISCO-quarters (offtake above contracted capacity, a real and verifiable occurrence), and falls well below it in others. Fifty-six per cent of observations sit below 0.95, where demand may bind; twenty-two per cent sit above 0.98, where supply clearly does. Nine of the eleven DISCOs cross this threshold in both directions at least once over the panel. The binding constraint genuinely switches (across DISCOs and within them over time), which is the precondition a disequilibrium model needs, and cannot be assumed to hold without checking.

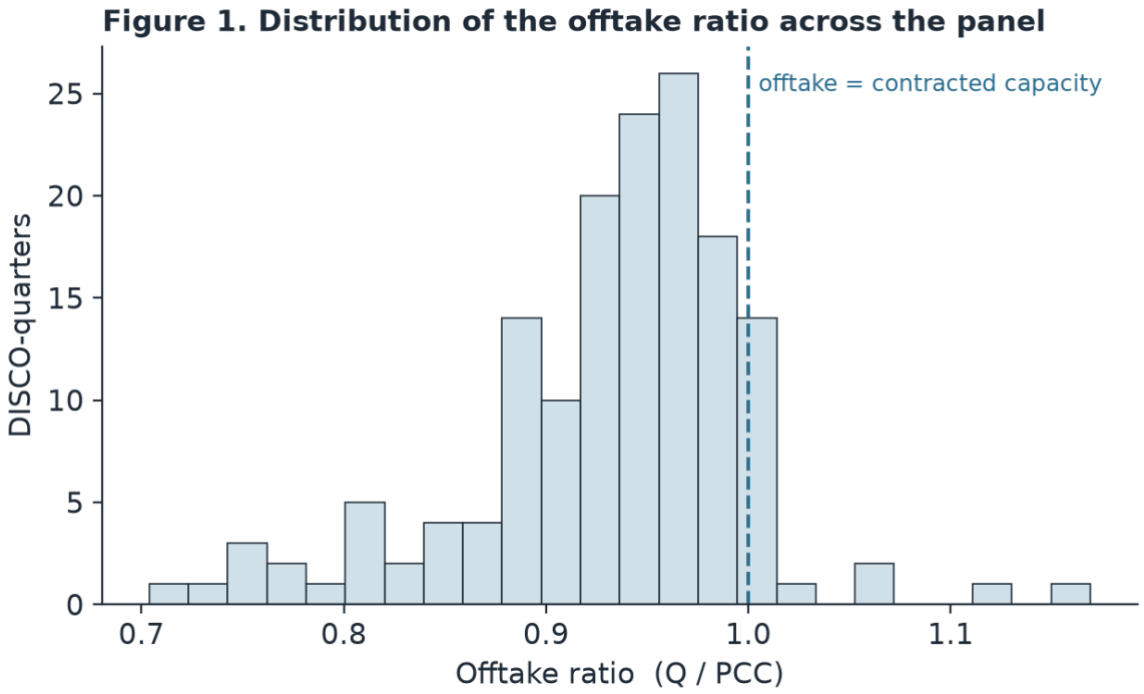


Figure 1. Distribution of the offtake ratio (Q/PCC) across the 154 DISCO-quarters. The dashed line marks the point at which offtake equals contracted capacity.

5.0 Model

5.1 Model Specification

Let i index the eleven DISCOs and t index quarters. Demand and supply are specified as separate linear equations:

$$D_{it} = \beta_0 + \beta_1 \ln(IGR_i) + \beta_2[\ln(IGR_i)\ X\ Lagos_i] + \beta_3 \cos\left(\frac{2\pi q_t}{4}\right) + \beta_4 \sin\left(\frac{2\pi q_t}{4}\right) + u_{1,it} \quad 1.1$$

$$S_{it} = \gamma_0 + \gamma_1 PCC_{it} + \gamma_2 PAF_t + \gamma_3 \cos\left(\frac{2\pi q_t}{4}\right) + \gamma_4 \sin\left(\frac{2\pi q_t}{4}\right) + \sum_i \delta_i DISCO_i + u_{2,it} \quad 1.2$$

with observed offtake given by

$$Q_{it} = \min(D_{it}, S_{it}) \quad 1.3$$

$D_{it}\ and\ S_{it}$ are never observed individually; only their minimum, Q_{it}, is. IGR_i is the DISCO's aggregated state Internally Generated Revenue; $Lagos_i$ is an indicator equal to one for the two Lagos-area DISCOs (Eko, Ikeja) and zero otherwise; $q_t \in \{1,2,3,4\}$is calendar quarter; PCC_{it} is contracted capacity; PAF_t is national plant availability, common to all DISCOs in a given quarter; $DISCO_i$ are DISCO fixed effects (ten dummies, Yola the reference category). The disturbances are assumed $u_{1,it} \sim N(0, \sigma_1^2)$, $u_{2,it} \sim N(0, \sigma_2^2)$ independent of each other and across i and t, the maintained assumption of Maddala and Nelson's (1974) Model 1. This assumption is tested directly in Section 6.

5.2 Identification

The model is identified by two exclusion restrictions, each grounded in how Nigeria's electricity market is actually administered rather than in a statistical convenience.

Contracted capacity and plant availability are excluded from the demand equation because both are supply-side quantities set independently of a DISCO's current desire to consume. Contracted capacity is fixed through NERC's regulatory tariff-order process and revised on an

administrative schedule, not in response to a DISCO's offtake in the same quarter (Section 3). Plant availability is a single national figure, common to every DISCO in a given quarter, reflecting the condition of generation plant rather than any individual DISCO's demand. Neither variable can plausibly be driven by how much electricity a DISCO's customers want in that quarter; both can plausibly constrain how much the DISCO receives.

State economic activity is excluded from the supply equation on the reverse logic: NERC's capacity allocation is set by regulatory order, not by a live assessment of a state's current tax revenue, and nothing in the allocation process adjusts a DISCO's contracted capacity because its franchise territory's Internally Generated Revenue rose or fell in a given quarter. The channel runs the other way: a wealthier, more commercially active franchise area is expected to want more electricity, not to be supplied more electricity for having a larger state budget.

One channel qualifies this restriction. Contracted capacity allocation is weighted, in part, by a DISCO's customer base (Section 3), and customer base and state economic activity are themselves correlated: a wealthier state plausibly has more electricity customers as well as higher IGR. If this correlation were left unaddressed, part of what looks like a demand response to IGR could instead be capturing a supply allocation that happens to be larger where IGR is also larger. The DISCO fixed effects in the supply equation absorb exactly this channel: they capture each DISCO's time-invariant baseline allocation, including whatever part of that allocation is explained by its customer base, leaving PCC's own coefficient to be identified from within-DISCO, quarter-to-quarter variation in contracted capacity, driven by regulatory revision rather than by a DISCO's underlying size. The same fixed effects that address this concern do not appear in the demand equation, where the corresponding cross-sectional variation in IGR is precisely what is needed to identify β_1.

Under these restrictions, the regime each DISCO-quarter occupies (demand-constrained or supply-constrained) is not assumed. It is recovered from the joint likelihood in Section 5.4, as the relative position of the fitted demand and supply curves at the observed quantity.

5.3 Functional Form

Two choices in the specification above depart from an equally natural alternative and are defended here.

Seasonality is modelled as a smooth annual cycle, not a separate coefficient per quarter. An earlier specification using three quarterly dummies (with Q4 as reference) left the demand equation's Q1 coefficient effectively unidentified: no DISCO-quarter in Q1, across the full panel, ever approaches its fitted demand ceiling, so the data contain no information about a Q1-specific demand shift distinct from the pattern traced by the other three quarters. The harmonic terms impose a single, smooth annual cycle instead of estimating each quarter independently, using two parameters rather than three and, critically, forcing Q1's implied position onto the curve traced by the quarters that do reveal information, rather than leaving it to float on data that cannot pin it down. The harmonic form assumes the underlying seasonal pattern is smooth rather than allowing an arbitrary shift at any single quarter. Section 6 reports the diagnostic that motivated this choice and confirms the resulting seasonal shape tracks Nigeria's rainy-to-dry seasonal cycle.

The IGR coefficient is allowed to differ for the two Lagos-area DISCOs (Eko, Ikeja). This interaction was adopted by comparing a pooled specification, in which every DISCO shares a single demand elasticity with respect to IGR, against one permitting the Lagos-area elasticity to differ; the latter is preferred on model-fit grounds (Section 6). The economic reading is that a DISCO already close to demand saturation in the country's largest and most developed commercial centre would be expected to show a flatter (though not necessarily weaker) relationship between further economic growth and additional electricity demand, since much of the connection between economic activity and grid demand may already have been made. Section 6 reports this comparison, and the confidence attached to the interaction term, directly.

5.4 Estimation

Because $D_{it}\ and\ S_{it}$ are independent and normal, the density of the observed minimum has a closed form. Writing $z_{1,it} = (Q_{it} - D_{it})/\sigma_1$ and $z_{2,it} = (Q_{it} - S_{it})/\sigma_2$ with $\emptyset$ and Φ the standard normal density and distribution functions:

$$f(Q_{it}) = \frac{1}{\sigma_1}\emptyset(z_{1,it})[1 - \Phi(z_{2,it})] + \frac{1}{\sigma_2}\emptyset(z_{2,it})[1 - \Phi(z_{1,it})] \quad 1.4$$

The first term is the likelihood contribution when Q_{it} equals demand and supply exceeds it; the second, when Q_{it} equals supply and demand exceeds it. The log-likelihood, summed over all 154 DISCO-quarters, is maximised jointly over $(\beta, \gamma, \sigma_1, \sigma_2)$.

This likelihood is not well behaved under naive numerical optimisation. Coefficients on regressors of very different scale (contracted capacity in the hundreds, seasonal terms near unity) distort a general-purpose optimiser's step sizes, and Maddala and Nelson's own multiple-maxima warning (Section 2.2) is not hypothetical for this model: independent runs from different starting points can converge to different, non-global optima. Three measures address this. Regressors are standardised before estimation and coefficients converted back to their original units afterward. The gradient of the log-likelihood is derived analytically rather than approximated numerically, and checked against a numerical gradient before use. Each specification is estimated from twelve independent starting points, and any run whose implied $\sigma_1 \; or \; \sigma_2$ collapses toward zero or diverges (a degenerate fit rather than a genuine optimum) is discarded; the reported estimate is the highest surviving log-likelihood, accepted only once multiple starting points converge to it independently.

Standard errors are computed from the numerical Hessian at the optimum, clustered by DISCO to account for repeated observations within each distribution company (Section 6).

6.0 Results

6.1 Demand Estimate

Table 3. Estimated demand, supply, and the suppressed-demand gap (panel averages).

Quantity	**GW**
Observed Offtake	3.43
Estimated demand ceiling ($\widehat{D}, existing\ connected\ network$)	5.34
Suppressed demand gap	1.91

Averaged across the panel, Nigeria's eleven DISCOs drew 3.43 GW against an estimated demand ceiling of 5.34 GW, 56 per cent above what was delivered. The gap is not evenly distributed: only three DISCOs (Abuja, Eko, Ikeja) ever approach their fitted demand ceiling closely enough to be classified as demand-constrained, a pattern detailed in Section 6.4.

6.2 Coefficient Estimates

Table 4. Demand equation coefficients, with cluster-robust standard errors.

Variable	**Coefficient**	**Cluster-robust SE**	**t**
Intercept	-3994.42	6.76	-590.60
ln(IGR)	173.05	4.08	42.39
ln(IGR) X Lagos	-2.90	0.11	-25.85
$\cos(2\pi q/4)$	17.47	1.15	15.22
$\sin(2\pi q/4)$	74.43	6.11	12.18

Table 5. Supply equation coefficients, with cluster-robust standard errors.

Variable	Coefficient	Cluster-robust SE	t
Intercept	23.72	21.75	1.09
PCC	0.736	0.095	7.72
PAF	-0.257	0.588	-0.44
$\cos(2\pi q/4)$	0.06	1.86	0.03
$\sin(2\pi q/4)$	1.73	1.52	1.14
Abuja	104.93	43.72	2.40
Benin	62.31	20.40	3.05
Eko	96.83	34.35	2.82
Enugu	48.25	19.10	2.53
Ibadan	80.42	31.12	2.58
Ikeja	114.32	43.36	2.64
Jos	15.05	9.03	1.67
Kaduna	19.94	11.19	1.78
Kano	24.01	11.37	2.11
Port Harcourt	45.94	15.89	2.89

DISCO effects are relative to Yola, the omitted reference category. σ_1 = 10.46, σ_2= 14.08. Log-likelihood: -621.96.

Ln(IGR) is positive and precisely estimated: a DISCO whose franchise territory's formal economic activity is higher draws more from the grid, holding season constant. The interaction term is negative and similarly precise: the Lagos-area DISCOs show a smaller IGR elasticity than the rest of the network, discussed in Section 6.3. On the supply side, contracted capacity converts to deliverable supply at close to unit elasticity (0.74); plant availability, a single figure common to all DISCOs each quarter, carries no reliable explanatory power once DISCO fixed effects are included, discussed in Section 6.4. Eight of the ten estimated DISCO effects are significant at the 5 per cent level; the remaining two (Jos, Kaduna) clear a 10 per cent threshold. All ten are positive, consistent with each DISCO carrying a real baseline supply allocation above Yola's.

This symmetry on the supply side is not mirrored on the demand side: as Section 6.4 reports, the observations that identify the demand equation's slope are concentrated in three of the eleven DISCOs, not spread evenly across the panel the way the supply-side fixed effects are.

6.3 Specification search

Demand-side variables. Six candidate variables were tested as drivers of cross-DISCO demand variation, each fit through the full estimation procedure rather than screened by simple correlation. Table 6 reports the outcome.

Table 6. Demand-side candidate variables tested.

Candidate	Tested	Outcome
Total Franchise Population	Alone	Significant, wrong-signed
Population, urban-masked	Alone	Significant, wrong-signed
Built-up surface area	Alone; with IGR	Not significant alone; unstable sign when combined with IGR
Built-up surface, residential-only	Alone	Statistically identical to total built-up area

Nighttime lights (VIIRS)	Alone; with IGR	A bit significant alone; sign reversed when combined with IGR
New customer connections (quarterly change)	Alone; with IGR	Not significant either specification
State IGR	Alone; with each of the aforementioned	Significant, correctly signed, stable across every combination

Population, in both raw and urban-masked form, is significant but wrong-signed: DISCOs covering larger, more rural franchise territories show lower estimated demand, the opposite of what a demand-scale variable should show. This is consistent with such territories combining low economic density with large land area rather than genuinely low demand. Built-up land area and its residential-only variant carry no independent signal once tested properly, and nighttime lights (significant on its own) reverse sign when paired with IGR, indicating the two are capturing overlapping variation rather than distinct information. None of the six candidates identifies a demand-side effect distinct from formal economic activity, a limit of what the available data can separately identify.

Group interactions. Two departures from a single, pooled demand elasticity were tested. Allowing the IGR elasticity to differ for the Lagos-area DISCOs (Eko, Ikeja) improved the Akaike Information Criterion by 12.4 points over the pooled specification (a substantial gain given the one additional parameter) and is retained in the reported specification (Table 4). Allowing the seasonal shape, rather than the IGR elasticity, to differ by DISCO grouping was tested twice: once for the same Lagos-area split, once for a north–south climate split. Neither improved the fit; the Lagos-area seasonal interaction left the log-likelihood unchanged to four decimal places, and the north–south interaction produced standard errors that could not be computed, indicating the additional parameters were not identified by the data. Both are excluded from the reported specification.

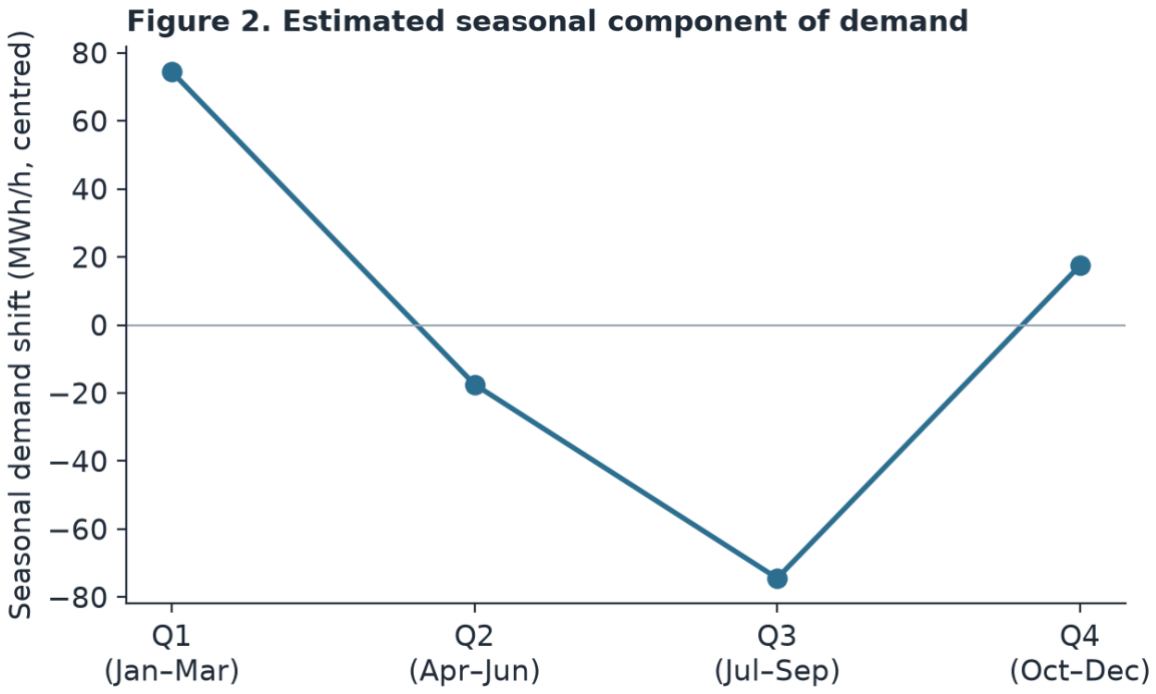


Figure 2. Estimated seasonal component of demand across the four quarters, centred on its mean.

6.4 Robustness

Reference year and time variation. The headline specification uses a single, static 2023 cross-section of IGR. Two questions arise: does the choice of reference year matter, and does using IGR's actual year-to-year movement instead of a fixed snapshot change the result? Neither does materially. Re-estimating with 2022 or 2024 in place of 2023 leaves the IGR coefficient, log-likelihood, and implied demand gap effectively unchanged (173.0–173.1 GW-scale coefficient across all three years). A fully time-varying specification, using each observation's own year's IGR rather than a fixed 2023 snapshot, produces a marginally better fit and a coefficient in the same range, at some cost to precision given only three years of usable annual variation. A further specification lagging the assigned IGR value by one year (on the reasoning that a given year's IGR may not be known to economic agents until partway through the following year) produced an unstable, wrong-signed estimate and is not reported as a reliable result; the instability persisted after expanding the underlying IGR series back to 2016, ruling out insufficient data as the cause.

The April 2024 tariff order. NERC's Band A tariff increase, effective April 2024, raised the possibility that the demand equation's independence from price-like shocks is violated for a subset of high-consumption customers. Adding a post-April-2024 indicator to the demand

equation, and separately an interaction between that indicator and IGR, produces no statistically detectable shift: the level-shift specification is preferred by neither the data nor the Akaike criterion over the specification without it, and the interaction specification could not be estimated reliably. The shock is not detectable at the DISCO-quarter level of aggregation used here, plausibly because it affects a customer class narrower than the aggregation captures, or because its effect was smaller or shorter-lived than initially assumed.

Standard errors. Clustering by DISCO (Table 4) tightens rather than widens the demand-side standard errors relative to the naive case. This is the expected outcome, not an anomaly, when residuals within a cluster are negatively rather than positively correlated over time, plausible here given that DISCOs move between demand- and supply-constrained regimes rather than sitting persistently in one, so a period of over-prediction is often followed by under-prediction within the same DISCO rather than by another over-prediction.

Regime composition, out-of-sample accuracy, and leave-one-out stability. Across the full 154-observation panel, 10 DISCO-quarters (6.5 per cent) are classified as demand-constrained ($\pi_demand > 0.5$); the remaining 144 are supply-constrained. These observations are concentrated in three DISCOs, Abuja, Eko, and Ikeja, and no others; every other DISCO in the panel is classified as supply-constrained in all fourteen quarters. The demand equation's identification therefore rests on a narrow slice of the data.

The out-of-sample test is consistent with this split at a smaller scale. The model was re-estimated on the first eleven quarters of the panel (2022Q4–2025Q2) and used to predict the final three, held out entirely from estimation. Predicted offtake differs from actual offtake by 7.6 per cent on average, 7.1 per cent among held-out supply-constrained observations (n=26) and 9.6 per cent among held-out demand-constrained observations (n=7), the latter figure resting on as few observations as the full-sample regime split would predict.

The consequence of this concentration was tested directly. The demand coefficients were re-estimated eleven times, each excluding one DISCO's data entirely, and fourteen times, each excluding one quarter; each refitted set of coefficients was then evaluated on the full, unchanged panel to isolate genuine coefficient sensitivity from the mechanical effect of removing a DISCO's own contribution to the national total. Excluding any of the eight DISCOs with no demand-constrained observations changes the resulting gap by at most 0.30 GW and the IGR coefficient by less than 2 per cent. Excluding any of the three that carry the panel's

demand-constrained observations changes both substantially: without Abuja, the IGR coefficient rises from 173 to 200 and the Lagos-area interaction nearly disappears (−2.9 to −0.4), and the gap falls to 0.52 GW; without Eko or Ikeja, the IGR coefficient rises to roughly 250 and the gap falls to 1.40–1.48 GW. The quarterly exclusions show the same pattern, concentrated in the quarters where these three DISCOs' demand-revealing observations occur. Across all twenty-five refits, the gap ranges from 0.52 to 2.21 GW (Appendix Table A1). The headline estimate is not broadly supported across the panel: it is anchored to three DISCOs' revealed behaviour.

6.5 The independence assumption

Model 1 assumes the demand and supply disturbances are independent. This assumption was tested directly by re-estimating the model with their correlation, ρ, as a free parameter rather than fixed at zero.

The estimate is uninformative. Every converged run returns ρ at or beyond -0.9999 (the edge of the range the optimiser was permitted to search) regardless of the range's width: widening the permissible bound from \pm 0.9999 to \pm 0.99999966 moved the estimate to the new edge rather than settling at an interior value, and excluding the panel's most extreme observations (the DISCO-quarters with offtake exceeding contracted capacity) produced the same boundary result. A parameter that consistently pins at whatever limit it is given, rather than settling within it, is not being estimated so much as it is failing to be identified: standard inference (the t-statistic and likelihood-ratio test that would ordinarily accompany a boundary estimate) does not apply when the true value may lie outside the region the data can distinguish from the boundary itself.

Because ρ cannot be estimated freely, its consequence for the headline figure was instead assessed directly, by fixing ρ at a small set of plausible values and re-estimating everything else. Table 7 reports the result. The response is asymmetric. Positive correlation moves the demand ceiling only marginally: at $\rho = 0.50$, $\hat{D}$ falls from 5.34 to 5.31 GW. Negative correlation moves it considerably more: at $\rho = -0.50$, $\hat{D}$ falls to 4.58 GW and the gap to 1.15 GW, a reduction of about two-fifths from the headline figure. This asymmetry is consistent with the direction the free estimate always moved toward: every unconstrained run of ρ pushed toward strongly negative values, never positive, so fixing ρ in that direction pulls the model toward

where it was already straining to go, while a positive value pulls against that tendency and the estimate barely shifts.

Table 7. Estimated demand ceiling and suppressed-demand gap under fixed values of shock correlation ρ

$\rho(fixed)$	$\widehat{D}(GW)$	Gap (GW)
-0.50	4.58	1.15
-0.25	4.55	1.12
0.00	5.34	1.91
+0.25	5.44	1.89
+0.50	5.31	1.88

The data cannot establish whether the assumption holds. What Table 7 does establish is the direction and scale of the risk if it does not: a violation in the negative direction, which is the direction the data themselves point toward when unconstrained, would lower the headline suppressed-demand gap by up to two-fifths. The reported model retains independence as a maintained assumption, and the 1.91 GW figure should be read as the estimate under that assumption, with Table 7 indicating how much it would move if the assumption fails in the direction the data suggest is more likely.

7.0 Discussion

7.1 What the estimate measures

The estimate is an average, not a peak. $\hat{D}$ = 5.34 GW is the mean, across the panel, of the demand the connected distribution network would draw at its prevailing structure of economic activity if generation and transmission did not bind, expressed in MWh/h, the same hourly-average unit in which contracted capacity is allocated. It is not an estimate of peak or instantaneous demand, and it does not describe the capacity a fully reliable system would require, which depends on the shape of demand within a day rather than its average level. The quarterly-average resolution of the underlying data does not support inference about that intra-quarter or intra-day shape.

The estimate is also bounded to the connected network. It reflects the demand of customers already served by the eleven DISCOs, recovered from their offtake behaviour; it excludes latent demand from unconnected households and firms, which by construction leave no trace in DISCO offtake and cannot be recovered from it. The 1.91 GW gap between $\hat{D}$ and observed delivery is therefore suppressed demand of the existing network (what current customers would draw if supplied without constraint), not a measure of total national electricity demand.

A further gap sits between $\hat{D}$ and what a paying customer's meter would show, even before this suppressed-demand adjustment is considered. DISCO offtake is demand at the point where a DISCO draws from the grid, not demand at the point of final consumption:

$$D_{grid} = D_{customer} + L \tag{1.5}$$

where L is the technical and commercial loss incurred between the grid intake point and the customer, reported nationally as ATC&C loss and running to several tens of per cent in a typical Nigerian DISCO-quarter. $\hat{D}$ therefore estimates the demand ceiling at the DISCO trading point; the demand ceiling actually reaching customers, net of L, is correspondingly smaller and is not separately estimated here.

7.2 Interpreting the coefficients

The demand side is driven by formal economic activity. Among the variables tested, only state IGR identifies cross-DISCO demand variation cleanly, and it does so strongly (Table 4): franchise territories with higher formal economic activity draw more power, holding season constant. That population and built-up land area do not survive as demand drivers, while a monetised measure of economic activity does, is consistent with Nigerian electricity demand at the distribution level being shaped more by the scale of formal commercial and industrial activity than by headcount or settlement footprint.

The Lagos-area interaction qualifies this relationship rather than overturning it. The two Lagos DISCOs show a smaller demand response to IGR than the rest of the network. A plausible reading is that in the country's most developed commercial centre, the link between incremental economic activity and incremental grid demand is closer to saturation than elsewhere, so further activity translates less directly into additional offtake. This coefficient is retained on model-fit grounds but is the least certain in the specification. Eko and Ikeja together account for six of the panel's ten demand-constrained observations, so the interaction is estimated largely from the same narrow slice of data that identifies the demand equation as a whole. It is best read as a qualification supported by a thin part of the data.

The supply side behaves as the institutional structure implies. Contracted capacity converts to deliverable supply at close to unit elasticity (0.74): a DISCO's administrative allocation is, to first order, what it receives. National plant availability adds no reliable explanatory power once DISCO fixed effects are present, as expected for a single national figure common to every DISCO in a quarter, which carries little information to distinguish one DISCO's supply from another's once each DISCO's baseline allocation is absorbed. The DISCO fixed effects themselves are uniformly positive and mostly significant, consistent with each distribution company holding a real, persistent supply allocation above that of the reference DISCO.

7.3 Limitations

Four limitations bound what these results establish.

First, the estimate covers the connected network only. Latent demand from unconnected households and firms is outside the model by construction (Section 7.1); the figure reported here is not, and should not be read as, total national demand.

Second, demand is identified by formal economic activity alone. Six candidate variables intended to capture household or settlement-based demand (population, urban-masked population, built-up area, residential built-up area, and new-connection flow) failed to identify a demand effect distinct from IGR (Section 6.3). This is a limit of what the available data separate, not evidence that household demand is unimportant; the reported demand equation should be understood as capturing the formal-economic-activity channel, with any distinct household channel unresolved.

Third, the independence of demand and supply shocks is assumed, not verified. A direct test was inconclusive: the correlation parameter was not identified by the sample (Section 6.5). The model retains independence as a maintained assumption.

Fourth, the panel is short, and the demand-constrained regime within it is thin. Fourteen quarters across eleven DISCOs is sufficient to estimate the reported specification and to support the robustness checks in Section 6, but only ten of the 154 observations, concentrated in three DISCOs, are classified as demand-constrained (Section 6.4). This limits what can be asked of the data: it is why the shock correlation cannot be identified (Section 6.5), why the time-varying IGR specification loses precision, why the out-of-sample test's demand-constrained subset is small, and why the headline gap moves substantially when any of these three DISCOs is excluded from estimation (Section 6.4). The national figure should accordingly be read alongside the DISCO-level results in Appendix Table A1, not in place of them; the estimate is best supported for Abuja, Eko, and Ikeja specifically, and least independently corroborated for the remaining eight DISCOs.

7.4 Relation to existing estimates

Against prior suppressed-demand estimates. Existing Nigerian estimates are not directly comparable in units or horizon (Ugbehe et al. (2025) report projected national energy in GWh per year to 2050; Ojameruaye (1988) worked in NEPA-era aggregates predating the DISCO structure entirely), so the contrast is one of method, not a common figure. Where those studies fix the correction from consumption to demand before estimation, the figure here is recovered from the joint behaviour of offtake and contracted capacity, with the constrained status of each observation identified by the model rather than assumed (Section 2.3).

Against observed delivery and capacity. The estimate is disciplined by the sector's own aggregates. Nigeria holds roughly 13 GW of installed generation capacity but delivered an average of about 4.5 GW to the grid in 2023 (IEA, 2024; African Development Bank, cited in AllAfrica, 2025). This paper's 3.43 GW average DISCO offtake, measured over a different window (2022Q4–2026Q1) and at a narrower point in the supply chain, is plausibly consistent with that figure rather than in tension with it: the IEA/ADB figure is measured at the point of generation and transmission, before transmission losses are subtracted and before offtake by large industrial customers who purchase directly from generators under Nigeria's "eligible customer" framework rather than through a DISCO. Both factors would be expected to widen the gap between total grid delivery and DISCO-metered offtake specifically, in the direction observed here. The estimated ceiling of 5.34 GW places connected-network suppressed demand at 56 per cent above delivery, well below installed capacity, and thus not a claim that the grid could serve its estimated demand today, though materially above what is currently delivered.

Against self-generation. The most telling external check is the scale of self-generation. Nigerian firms and households operate an estimated 8–14 GW of decentralised diesel and gas generating capacity (by several accounts equal to or exceeding grid-connected capacity), and roughly 40 per cent of electricity consumed nationally is self-generated (IEA, 2024; ESI-Africa, 2024; Climate Technology Centre & Network, 2017). This figure is not directly comparable to the estimate here and should not be read as one: self-generation reflects total unmet demand across both connected and unconnected users, and is driven substantially by demand for reliability and continuity that an average-power figure does not capture. But its sheer scale (a captive fleet on the order of the entire grid) is qualitatively consistent with the central finding that grid offtake substantially understates what connected customers would draw if supplied, and locates the 1.91 GW connected-network gap as a conservative lower bound on the broader unmet demand the self-generation market reveals.

8.0 Conclusion

This paper estimated the suppressed electricity demand of Nigeria's connected distribution network by treating observed offtake as the short side of a market that does not clear, and recovering demand and supply as separately identified equations rather than assuming the correction between consumption and demand in advance. The existing network is estimated to draw 5.34 GW on average against 3.43 GW delivered (a suppressed demand gap of 1.91 GW, or 56 per cent above current delivery), with formal economic activity (rather than population or settlement footprint) as the operative demand driver, and contracted capacity converting to supply at close to unit pass-through. The estimate survives out-of-sample testing, but it is anchored to the three DISCOs whose behaviour reveals demand (Abuja, Eko and Ikeja) and assumes independent demand and supply shocks; Sections 6.4 and 6.5 report how far it moves when either condition is relaxed.

The contribution is as much methodological as empirical. The disequilibrium framework has existed for fifty years and been applied across rationed markets, but not, in its full joint-estimation form, to electricity demand in a developing-country grid. Doing so removes the central weakness of existing suppressed-demand estimates for such settings (that the size of the demand-supply gap is fixed by assumption before the data are consulted) and replaces it with a figure the data identify directly, including which observations were demand- and which supply-constrained. The method's requirements are modest: quarterly offtake, an administrative supply allocation, and a demand shifter excluded from supply. Many rationed power systems report exactly this.

That points to the wider application. The measurement problem this paper addresses (quantity rationed away from what price would clear, so that observed consumption reveals only the short side) is not specific to chronic under-supply. It has re-emerged, since 2024, in the most developed electricity markets, where data-centre and AI load has outrun the grid's physical capacity to connect it: at the end of 2025, US interconnection queues held roughly 2,061 GW of pending capacity, around one and a half times the installed base, with multi-year waits to connect. There, price clears the wholesale market but not the queue for connection itself: a market with its own latent demand, its own constrained supply, and only their minimum observed. Estimating that gap is the same problem in a different setting, and one this framework is suited to, though it is left here as a direction rather than a result. The method was built for

markets where price cannot do the work of allocation; such markets are becoming more common, not less.

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APPENDIX

Table A1. Leave-one-out sensitivity of the estimated gap.

Excluded	**Gap (GW)**	**Δ *from headline***
Abuja	0.52	-1.39
Eko	1.48	-0.43
Ikeja	1.40	-0.51
Benin	1.93	+0.02
Enugu	1.91	+0.00
Ibadan	2.14	+0.23
Jos	1.90	-0.01
Kaduna	1.92	+0.01
Kano	1.91	-0.00
Port Harcourt	2.21	+0.30
Yola	1.91	+0.00

Quarterly exclusions omitted for brevity; range 1.12–2.12 GW, largest deviations at 2022Q4, 2025Q4, and 2026Q1 (Section 6.4). Full table available on request or in the project repository.